

Multiple Choice

1. **Answer:** A

Options: A) real; B) imaginary; C) degenerate; D) linear

Note: Correct option: A - real

2. **Answer:** A

Options: A) decreased by a factor of 9; B) decreased by a factor of 49; C) decreased by a factor of 81; D) increased by a factor of 9

Note: Correct option: A - decreased by a factor of 9

3. **Answer:** D

Options: A) $(1/2) k T$; B) kT ; C) $(3/2) k T$; D) $3 kT$

Note: Correct option: D - $3 kT$

4. **Answer:** B

Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic

Note: Correct option: B - Constant volume

5. **Answer:** D

Options: A) 44 Hz; B) 196 Hz; C) 262 Hz; D) 393 Hz

Note: Correct option: D - 393 Hz

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '5'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '450 m'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The entropy of the system and its environment remains unchanged.'

Long Form Response

11. **Answer:** 10 eV. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Planck's constant. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key